

Biology Playbook

Leaf · Flower · Cell · Digestion · Respiration · Circulation · Health · Adaptation

ICSE Class 6 · Full Book · Final Exam Ready

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Chapter names and order can differ slightly between schools and publishers. Follow your school's exam syllabus.

How to use this playbook

Unit	Chapter
Plant life	1. The Leaf · 2. The Flower
Cells and the human body	3. The Cell · 4. Digestive System · 5. Respiratory System · 6. Circulatory System
Health and the environment	7. Health and Hygiene · 8. Habitat and Adaptation

⭐ **Exam Tip: How to score full marks in Biology**

1. **Diagrams carry big marks**: practise the leaf, flower, cell, digestive system, lungs and heart. Use a pencil, draw label lines straight with a ruler, and **don't cross label lines**.
2. Learn the **functions** in pairs: "part → its job".
3. "Differentiate" questions: always answer in a **table**.

1. The Leaf

- The **leaf** is the **food factory** of the plant. It is green because it contains **chlorophyll**.

Parts of a leaf

Part	Description
Leaf base	Attaches the leaf to the stem
Petiole	The stalk of the leaf
Lamina (leaf blade)	The flat green part where food is made
Midrib	The main central vein
Veins	Carry water in and food out, and give support
Margin	The edge of the lamina
Apex	The tip of the leaf

- A leaf with a petiole is **petiolate**. Without a petiole it is **sessile**.

Venation (the pattern of veins)

Reticulate venation	Parallel venation
Veins form a **net**	Veins run **parallel** to each other
Found in **dicots**	Found in **monocots**
Mango, peepal, hibiscus	Banana, grass, maize, wheat

Simple and compound leaves

- **Simple leaf:** one undivided lamina (mango, guava)
- **Compound leaf:** the lamina is divided into **leaflets** (neem, rose, tamarind)

Functions of the leaf

1. **Photosynthesis**: making food
2. **Transpiration**: losing extra water as vapour through **stomata**
3. **Gas exchange** through stomata
4. **Storage** (onion, aloe vera), **protection** (cactus spines are modified leaves), **climbing** (pea tendrils)

Photosynthesis

Carbon dioxide + water —(sunlight, chlorophyll)→ glucose (food) + oxygen

- **Raw materials:** CO_2 (from the air, through stomata) and water (from the soil, through the roots)
- **Conditions: sunlight** and **chlorophyll**
- **Products:** glucose (stored as **starch**) and **oxygen** (released)
- **Starch test:** iodine solution turns starch **blue-black**.

Stomata

- Tiny pores, mostly on the **lower surface** of the leaf, each surrounded by two **guard cells** that open and close it.

🧠 Memory Trick

Photosynthesis = "Photo" (light) + "synthesis" (making): making food with light. Leaf parts from bottom to top: "Bina Pet Lamba Mota Aam": Base, Petiole, Lamina, Midrib, Apex.

📍 Mumbai Connection

Sanjay Gandhi National Park in Borivali, and the mangroves along Mumbai's coast, produce huge amounts of oxygen by photosynthesis. They are called the **"lungs of Mumbai"**!

🎲 Play It in Class: Leaf Detective Walk

Students collect 5 fallen leaves from the school garden. They classify each one: simple or compound? reticulate or parallel? petiolate or sessile? Then they make a leaf chart with labels.

✏️ Quick Check

1. Why is the leaf called the food factory of the plant?
2. Write the word equation for photosynthesis.
3. Give two examples each of reticulate and parallel venation.
4. What are stomata? Name the cells that surround them.

2. The Flower

- The **flower** is the **reproductive part** of the plant. It develops into **fruits and seeds**.

Parts of a Flower (from outside to inside)

Whorl	Parts	Function
Calyx	Sepals (green)	Protects the bud
Corolla	Petals (colourful)	Attracts insects for pollination
Androecium (male)	**Stamens**: an **anther** (makes **pollen grains**) + a **filament** (stalk)	Makes pollen
Gynoecium (female)	**Pistil**: a **stigma** (sticky top) + a **style** (tube) + an **ovary** (contains **ovules**)	Makes ovules, which become seeds

- **Pedicel:** the stalk of the flower. **Thalamus:** the swollen base the flower parts sit on.
- **Bisexual flower:** has both stamens and a pistil (hibiscus, mustard). **Unisexual flower:** has only one of them (papaya, cucumber, maize).

Pollination

- **Pollination** is the transfer of **pollen grains** from the **anther** to the **stigma**.
- **Self-pollination:** in the same flower or the same plant. **Cross-pollination:** between different plants of the same kind.
- **Agents:** insects (bees, butterflies), wind, water, birds and animals

Fertilisation and Fruit Formation

- The male cell (from pollen) **fuses** with the female cell (in the ovule): this is **fertilisation**.
- After fertilisation: the **ovary → fruit** and the **ovules → seeds**.

🧠 **Memory Trick**

From outside to inside: "Can Cats Ask Grandma?": **C**alyx, **C**orolla, **A**ndroecium, **G**ynoecium. **"Anther = Pollen Factory, Stigma = Sticky Landing Pad" Ovary becomes the fruit (O → F). Ovules become the seeds.**

📍 **Mumbai Connection**

Mumbai's famous **Alphonso (Hapus) mango** from Ratnagiri grows from a fertilised mango flower. The ovary becomes the juicy fruit, and the ovule becomes the seed (the "guthli")!

🎲 **Play It in Class: Flower Dissection**

Bring a hibiscus (shoe flower). Teams carefully remove each whorl and stick them on paper in order: sepals → petals → stamens → pistil. Label each one. Look at the pollen with a hand lens!

✏️ Quick Check

1. Name the four whorls of a flower.
2. What is pollination? Name two agents of pollination.
3. What do the ovary and the ovules become after fertilisation?
4. Give one example each of a bisexual and a unisexual flower.

3. The Cell: Structure and Functions

- The **cell** is the **structural and functional unit of life**. All living things are made of cells.
- **Robert Hooke** (1665) first saw cells in a thin slice of **cork** using his microscope.
- **Unicellular** organisms have one cell (amoeba, paramecium, bacteria). **Multicellular** organisms have many cells (humans, plants).
- **Largest cell:** the ostrich egg. **Longest cell in humans:** the nerve cell. **Smallest:** bacteria-like cells (mycoplasma).

Parts of a cell

Part	Function
Cell membrane	A thin outer layer that controls what enters and leaves
Cell wall (plants only)	A rigid outer layer made of **cellulose** that gives shape and protection
Cytoplasm	A jelly-like fluid where the organelles float
Nucleus	The **control centre** of the cell, containing the **chromosomes** (DNA)
Mitochondria	The **powerhouse**, which releases energy from food
Chloroplasts (plants only)	Contain chlorophyll, where photosynthesis happens
Vacuole	Stores water and wastes (large in plant cells, small or absent in animal cells)

Plant cell vs animal cell

Plant cell	Animal cell
Has a **cell wall**	No cell wall
Has **chloroplasts**	No chloroplasts
One large vacuole	Small vacuoles, or none
Usually **rectangular** (fixed shape)	Usually **round or irregular**

Levels of organisation

Cell → tissue → organ → organ system → organism Example: muscle cells → muscle tissue → heart → circulatory system → human

> 🧠 **Memory Trick**
> **The cell is a school!** Nucleus = **Principal** (controls everything). Mitochondria = **Canteen** (gives energy). Cell membrane = **Security guard** (decides who comes in). Cell wall = **Boundary wall**. Cytoplasm = **Playground** where everyone moves around. Vacuole = **Storeroom**.

🎲 Play It in Class: Build-a-Cell

Each team makes a cell model on a paper plate: green peas for chloroplasts, a cherry or bead for the nucleus, a rubber band for the membrane, a cardboard edge for the cell wall. Is it a plant cell or an animal cell? Other teams have to guess!

✏️ Quick Check

1. Who discovered the cell, and in what?
2. Which part is called (a) the powerhouse (b) the control centre of the cell?
3. Give three differences between plant and animal cells.
4. Write the levels of organisation in order.

4. The Digestive System

- **Digestion** is the breakdown of **complex food** into **simple, soluble** substances that the body can absorb.

The path of food (alimentary canal)

Mouth → oesophagus (food pipe) → stomach → small intestine → large intestine → rectum → anus

Organ	What happens there
Mouth	Teeth chew the food. **Saliva** (from the salivary glands) contains an enzyme that starts digesting **starch**. The tongue mixes the food.
Oesophagus	Food moves down by wave-like muscle movements (**peristalsis**).
Stomach	**Gastric juice**: **hydrochloric acid** (kills germs) + enzymes that digest **proteins**.
Small intestine	The **longest part** (about 6–7 m). **Bile** (from the liver) breaks fats into small drops, and **pancreatic juice** and **intestinal juice** complete digestion. **Villi** (finger-like projections) **absorb** the digested food into the blood.
Large intestine	Absorbs **water** and salts.
Rectum and anus	Undigested waste is stored and then removed (**egestion**).

Digestive glands

- **Salivary glands** (saliva), **liver** (bile, the **largest gland** in the body), **pancreas** (pancreatic juice)

Teeth

Type	Number (adult)	Job
Incisors	8	Cutting and biting
Canines	4	Tearing
Premolars	8	Chewing and grinding
Molars	12	Chewing and grinding
Total	**32**	Milk teeth: **20**

🧠 **Memory Trick**

The food's journey: "My Old Sister Sells Lovely Rasgullas Always": Mouth, Oesophagus, Stomach, Small intestine, Large intestine, Rectum, Anus. **Teeth: "I Can Play Music"**: Incisors (cut), Canines (tear), Premolars and Molars (grind).

📍 Mumbai Connection

When you eat a **vada pav**: the pav (starch) starts digesting in the **mouth** with saliva. The besan coating (protein) is digested in the **stomach** and small intestine. The oil (fat) is broken down by **bile** in the **small intestine**!

🎲 Play It in Class: Digestion Relay

Lay out stations around the classroom labelled with the organs. A student is the "food" and must visit them in order, saying what happens at each one. The team that finishes the journey fastest and correctly wins.

✏️ Quick Check

1. Define digestion.
2. Write the parts of the alimentary canal in order.
3. Name the largest gland of the body and its juice.
4. What are villi? What is their function?

5. The Respiratory System

- **Breathing:** taking in air (**inhalation**) and giving out air (**exhalation**).
- **Respiration:** the breakdown of food (glucose) **inside cells** to release **energy**. **Glucose + oxygen → carbon dioxide + water + energy**

The path of air

Nostrils → nasal cavity → pharynx → larynx (voice box) → trachea (windpipe) → bronchi → bronchioles → alveoli (air sacs) in the lungs

Part	Function
Nose	Hairs and mucus **filter** the dust. The air is warmed and moistened.
Trachea	**Rings of cartilage** keep it open.
Lungs	Two spongy organs in the chest, protected by the **rib cage**
Alveoli	Tiny air sacs where **oxygen enters the blood** and **CO_2 leaves the blood**
Diaphragm	A dome-shaped muscle below the lungs that helps breathing

The breathing mechanism

Inhalation	Exhalation
The diaphragm moves **down** (flattens)	The diaphragm moves **up** (dome shape)
The ribs move **up and out**	The ribs move **down and in**
The chest gets **bigger**, so air rushes **in**	The chest gets **smaller**, so air is pushed **out**

- **Breathing in through the nose is better than through the mouth**, because the nose filters, warms and moistens the air.
- An adult breathes about **15–18 times a minute** at rest.

🧠 **Memory Trick**

Air's path: "No Nonsense, People Like To Breathe Better Always": Nostrils, Nasal cavity, Pharynx, Larynx, Trachea, Bronchi, Bronchioles, Alveoli. **Breathing = mechanical (lungs). Respiration = chemical (cells).**

📍 **Mumbai Connection**

On high-AQI days in Mumbai, doctors advise wearing a **mask** outdoors. Dust and smoke harm the **alveoli** and can cause asthma. Trees on the roadside help to trap the dust.

🎲 **Play It in Class: Balloon Lungs Model**

Make a model with a plastic bottle (the chest), a balloon inside (the lung) and a rubber sheet at the bottom (the diaphragm). Pull the sheet down and the balloon fills. Push it up and the balloon shrinks. Students explain what they see.

✏️ Quick Check

1. Differentiate between breathing and respiration.
2. Write the path of air from the nostrils to the alveoli.
3. What is the role of the diaphragm?
4. Why should we breathe through the nose?

6. The Circulatory System

- The **circulatory system** carries **oxygen, food, water and hormones** to all the cells, and carries **wastes and CO_2** away.
- Its parts are the **heart**, the **blood vessels** and the **blood**.

The heart

- A muscular organ about the size of your **fist**, slightly to the **left** of the chest.
- It has **four chambers**: 2 upper **atria** (they receive blood) and 2 lower **ventricles** (they pump blood out).
- **Valves** stop blood from flowing backwards.
- The heart beats about **72 times per minute**. A **stethoscope** is used to hear the heartbeat.

Blood vessels

Arteries	Veins	Capillaries
Carry blood **away** from the heart	Carry blood **to** the heart	Very thin vessels that connect arteries to veins
Usually **oxygen-rich** blood	Usually **oxygen-poor** blood	Where the exchange with cells happens
Thick, elastic walls	Thin walls, with valves	One cell thick

- **Exception:** the pulmonary artery carries oxygen-poor blood to the lungs, and the pulmonary vein carries oxygen-rich blood back.

Blood

Part	Job
Plasma (a yellowish liquid, about 55%)	Carries food, wastes and hormones
Red blood cells (RBCs)	Contain **haemoglobin** and carry **oxygen**. They make blood red.
White blood cells (WBCs)	**Fight germs** (the body's soldiers)
Platelets	Help blood **clot** at a cut

- The **pulse** is felt at the wrist. It is the same as the heart rate.

> 🧠 **Memory Trick**
> **A for Arteries, A for Away from the heart. RBC = Red Bus Carrying oxygen. WBC = White Brave Cops fighting germs. Platelets = Plumbers fixing leaks (cuts)!**

🎲 Play It in Class: Pulse Party

Students find their pulse at the wrist and count it for 15 seconds × 4. Then they do 20 jumping jacks and count again. Compare: why does it increase? Make a class bar graph of resting and active pulse rates.

✏️ Quick Check

1. How many chambers does the human heart have? Name them.
2. Give two differences between arteries and veins.
3. What is the function of (a) RBCs (b) platelets?
4. What is a stethoscope used for?

7. Health and Hygiene

- **Health** is a state of complete **physical, mental and social well-being**, not just the absence of disease (World Health Organization).
- **Hygiene** means the practices that keep us healthy and clean.

Personal hygiene

Bathe daily, brush twice a day, cut your nails, wash your hands before eating and after using the toilet, wear clean clothes, sleep 8–10 hours, exercise.

Community hygiene

Proper garbage disposal (wet and dry dustbins), clean drains, no open defecation, no spitting, clean water supply, **don't let water collect** (it breeds mosquitoes).

Diseases

Type	How they spread	Examples
Communicable (infectious)	Germs pass from person to person through air, water, food, insects or contact	Cold, flu, TB, cholera, typhoid, malaria, dengue, COVID-19
Non-communicable	Don't spread from person to person	Diabetes, heart disease, cancer, deficiency diseases

Disease	Germ or cause	Spread by
Malaria	Protozoan (Plasmodium)	Female **Anopheles** mosquito
Dengue	Virus	**Aedes** mosquito (bites in the daytime, breeds in clean stagnant water)
Cholera, typhoid	Bacteria	Contaminated food and water
Tuberculosis (TB)	Bacteria	Air (coughing, sneezing)
Common cold	Virus	Air and contact

Balanced diet

- A **balanced diet** contains carbohydrates, proteins, fats, vitamins, minerals, fibre (roughage) and water **in the right amounts**.
- **Deficiency diseases:** scurvy (lack of vitamin C), night blindness (vitamin A), rickets (vitamin D), anaemia (iron), goitre (iodine).

Vaccination

- **Vaccines** protect us from diseases such as polio, measles, tetanus, hepatitis B and BCG (for TB).

🧠 Memory Trick

"Aedes in the Afternoon (daytime), Anopheles At night." Dengue mosquitoes bite in the daytime, malaria mosquitoes at night. **Vitamins and their diseases: "A–Night, C–Scurvy, D–Rickets"** (A night sky, C sea-scurvy pirates, D for "damaged bones").

📍 Mumbai Connection

During the **monsoon**, cases of **dengue, malaria and leptospirosis** rise in Mumbai because water collects in tyres, flower pots, construction sites and on terraces. The BMC runs "dry day" drives. Students can check their homes every Sunday for stagnant water!

🎲 Play It in Class: Germ Glitter Game

Put a little glitter (the "germs") on one student's hands. They shake hands with 3 friends. Watch the glitter spread! Then wash with water only, and then with soap. Which removes more? That's why we wash our hands with soap for 20 seconds.

✏️ Quick Check

1. Define health.
2. Differentiate between communicable and non-communicable diseases, with examples.
3. Name the mosquito that spreads (a) malaria (b) dengue.
4. Name the deficiency disease caused by a lack of vitamin C and of iodine.

8. Habitat and Adaptation

- **Habitat:** the natural home of a living thing, where it gets food, water, shelter and suitable conditions.
- **Adaptation:** special features that help an organism **survive** in its habitat.
- **Biotic components:** living things (plants, animals, microbes). **Abiotic components:** non-living things (air, water, soil, sunlight, temperature).

Types of habitats and adaptations

Habitat	Example	Adaptations
Desert (hot and dry)	**Camel**	Long legs (stay away from the hot sand), wide padded feet (don't sink), stores fat in its **hump**, rarely sweats, can go without water for days
	Cactus	Leaves become **spines** (less water loss), a thick green stem does photosynthesis and stores water, a thick waxy coating, deep roots
Polar (very cold)	**Polar bear**	Thick white fur (warmth and camouflage), a thick fat layer, a strong sense of smell
	Penguin	Thick skin and fat, huddles in groups, webbed feet for swimming
Aquatic (water)	**Fish**	A streamlined body, **gills** for breathing, fins and a tail, slippery scales
	Lotus, water lily	Floating broad leaves with **stomata on the upper surface**, hollow stems, a waxy coating
Mountain (cold)	**Yak, mountain goat**	Thick fur, strong hooves for climbing
	Pine, deodar trees	**Cone-shaped**, so snow slides off, with needle-like leaves
Grassland	**Lion, deer**	A lion's colour helps it hide in dry grass. A deer has long ears and eyes on the sides to watch for danger, and strong legs to run fast.
Forest (tree-dwelling)	**Monkey**	A long tail and gripping hands for climbing
Air (flying)	**Birds**	**Hollow light bones**, feathers, wings, a streamlined body

Plant adaptations

- **Climbers:** use tendrils (pea, grapes)
- **Insectivorous plants** (pitcher plant, Venus flytrap): catch insects to get the **nitrogen** missing from their soil
- **Mangroves:** breathing roots (**pneumatophores**) that grow **above** the salty, muddy ground

🧠 Memory Trick

"Every animal has a superpower for its home!" Camel = water tank (fat in the hump), fish = built-in snorkel (gills), polar bear = winter jacket (fur + fat), birds = lightweight backpack (hollow bones).

📍 Mumbai Connection

Mumbai's **mangroves** (at Mahim, Vikhroli and Airoli) have **breathing roots** that poke up from the salty mud, and they protect the city from floods and storms. Mumbai also has **flamingos** at Sewri and Thane Creek every winter. Their long legs help them wade, and their bent beaks filter food from the mud!

🎲 Play It in Class: Design-a-Creature

Teams get a habitat card (desert, polar, ocean, mangrove…). They invent a new animal with at least 4 adaptations, draw it and present it. The class votes for the most clever survivor!

✏️ Quick Check

1. Define habitat and adaptation.
2. Write three adaptations of a camel.
3. How are the leaves of a lotus adapted to life in water?
4. Why do mangroves have breathing roots?

Full-Book Revision Test

Section A (1 mark each)

1. Name the green pigment in leaves.
2. What does the ovary become after fertilisation?
3. Which organelle is called the powerhouse of the cell?
4. Where is bile produced?
5. Name the air sacs in the lungs.
6. Which blood cells fight germs?

Section B (2–3 marks each)

7. Draw and label the parts of a leaf.
8. Differentiate between self-pollination and cross-pollination.
9. Draw a plant cell and label five parts.
10. Name the four types of teeth and their functions.
11. Describe what happens to the diaphragm and ribs during inhalation.
12. Give two differences between arteries and veins.

Section C (4 marks each)

13. Explain photosynthesis: its raw materials, conditions and products.
14. Describe the digestion of food in the stomach and small intestine.
15. Explain three adaptations each of a cactus and a fish.

Answer Key

Ch 1: (1) It makes food for the whole plant by photosynthesis. (2) Carbon dioxide + water → (sunlight, chlorophyll) → glucose + oxygen (3) Reticulate: mango, peepal. Parallel: banana, grass. (4) Tiny pores on the leaf for gas exchange and transpiration. They are surrounded by guard cells.

Ch 2: (1) Calyx, corolla, androecium, gynoecium (2) The transfer of pollen from the anther to the stigma. Agents: insects and wind. (3) The ovary becomes the fruit, and the ovules become seeds. (4) Bisexual: hibiscus. Unisexual: papaya.

Ch 3: (1) Robert Hooke, in a slice of cork (1665) (2) (a) Mitochondria (b) Nucleus (3) Plant cells have a cell wall, chloroplasts and a large vacuole. Animal cells have none of these. (4) Cell → tissue → organ → organ system → organism

Ch 4: (1) The breakdown of complex food into simple, soluble substances the body can absorb. (2) Mouth → oesophagus → stomach → small intestine → large intestine → rectum → anus (3) The liver, which makes bile (4) Finger-like projections in the small intestine that absorb digested food.

Ch 5: (1) Breathing is the mechanical intake and output of air (lungs). Respiration is the chemical breakdown of glucose to release energy (cells). (2) Nostrils → nasal cavity → pharynx → larynx → trachea → bronchi → bronchioles → alveoli (3) It moves down to enlarge the chest (inhalation) and up to push air out (exhalation). (4) The nose filters, warms and moistens the air.

Ch 6: (1) Four: right and left atria, right and left ventricles (2) Arteries carry blood away from the heart and have thick walls. Veins carry blood to the heart, have thin walls and have valves. (3) (a) RBCs carry oxygen (b) Platelets help blood clot (4) To hear the heartbeat.

Ch 7: (1) A state of complete physical, mental and social well-being. (2) Communicable diseases spread from person to person (TB, cholera). Non-communicable diseases don't (diabetes, heart disease). (3) (a) The female Anopheles (b) Aedes (4) Scurvy (vitamin C) and goitre (iodine)

Ch 8: (1) Habitat: the natural home of an organism. Adaptation: features that help it survive there. (2) Hump fat, long legs, padded feet, rarely sweats (any three) (3) Broad floating leaves with stomata on the upper surface and a waxy coating. (4) The mud is waterlogged and has little oxygen, so the roots grow upwards to take in air.

Revision test: (1) Chlorophyll (2) The fruit (3) Mitochondria (4) In the liver (5) Alveoli (6) White blood cells (7) Leaf base, petiole, lamina, midrib, veins, margin, apex (8) Self-pollination happens within the same flower or plant. Cross-pollination happens between different plants of the same kind. (9) Cell wall, cell membrane, cytoplasm, nucleus, chloroplast, vacuole (10) Incisors cut, canines tear, premolars and molars grind. (11) The diaphragm moves down and the ribs move up and out, so the chest expands and air rushes in. (12) Arteries carry blood away from the heart and have thick walls. Veins carry blood to the heart, have thin walls and have valves. (13) Raw materials: CO_2 and water. Conditions: sunlight and chlorophyll. Products: glucose (stored as starch) and oxygen. (14) Stomach: HCl kills germs and enzymes digest proteins. Small intestine: bile breaks up fats, and pancreatic and intestinal juices complete the digestion of carbohydrates, proteins and fats. Villi absorb the food. (15) Cactus: spines, a fleshy stem that stores water, a waxy coat, deep roots. Fish: a streamlined body, gills, fins, scales.